Quantum-Enhanced Deep Reinforcement Learning for Central Bank Digital Currency (CBDC) Liquidity Management

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Abstract

The introduction of a Central Bank Digital Currency (CBDC), such as the Digital Euro, presents a critical challenge to commercial bank liquidity management due to the potential for large-scale deposit conversion. This paper proposes a novel Quantum-Enhanced Deep Reinforcement Learning (QDRL) framework to optimize commercial bank funding strategies under CBDC-induced liquidity stress.

We address the barren plateau problem through theoretical analysis and empirical gradient variance measurements, demonstrating that our 4-qubit, 6-layer Hardware-Efficient Ansatz remains trainable throughout the 500,000 timestep training run via identity block initialization and layer-wise training. A controlled ablation study comparing the VQC to a classical neural network of equivalent parameter count reveals that quantum effects contribute meaningfully to performance enhancement, though the advantage is contingent on noise-free or noise-mitigated execution.

The simulation environment has been expanded from a single bank to a network of five heterogeneous commercial banks with interbank lending relationships, varying balance sheet compositions, and contagion dynamics. Under realistic NISQ noise models calibrated to IBM Quantum hardware, Zero Noise Extrapolation recovers meaningful signal, though performance degrades proportionally to gate error rates.

Results across four calibrated stress scenarios demonstrate that the Quantum-Enhanced SAC (QSAC) agent achieves superior performance, reducing average funding costs by 8.1% (95% CI: [6.2%, 9.9%]) compared to classical DRL baselines and eliminating Liquidity Coverage Ratio (LCR) breaches in out-of-sample simulation. However, when compared against Model Predictive Control baselines, the advantage narrows to 3.2% (95% CI: [1.1%, 5.3%]).

Current Limitations: This work represents a proof-of-concept hybrid quantum-classical framework tested exclusively in simulation. Results are contingent on (1) noise-free or noise-mitigated quantum hardware not yet commercially available at scale, (2) a stylized multi-bank environment that does not capture all complexities of real interbank markets, and (3) VQC expressibility limited to 4 qubits due to NISQ constraints. The economic estimates presented are sensitivity-based projections with wide confidence intervals, not validated empirical findings.

Keywords: Quantum Machine Learning, Deep Reinforcement Learning, Central Bank Digital Currency, Liquidity Management, Soft Actor-Critic, Variational Quantum Circuit, Barren Plateau, NISQ Computing, Systemic Risk, Quantum Explainable AI

1. Introduction

The European Central Bank (ECB) is actively progressing the Digital Euro project, recognizing the need for a resilient, efficient, and sovereign digital monetary system[1]. While a CBDC offers numerous benefits including payment efficiency and financial inclusion, it introduces a significant systemic risk: the potential for large-scale, rapid conversion of commercial bank deposits into risk-free CBDC, leading to liquidity stress in the banking sector[2]. The ECB estimates that under extreme scenarios, this drain could reach up to EUR 700 billion[3].

Managing this liquidity risk is a complex, high-dimensional, sequential optimization problem. Commercial banks must dynamically adjust their funding mix (interbank loans, asset sales, central bank facilities) to meet potential outflows while minimizing the cost of funding. Traditional optimization models, relying on classical assumptions and linear programming, struggle to adapt to the non-linear, stochastic nature of CBDC adoption and market volatility[4].

Deep Reinforcement Learning (DRL) offers a powerful alternative, as it is designed for optimal sequential decision-making in dynamic environments. However, the complexity of the problem is significant. The large state-space (market rates, bank balance sheet, CBDC flow) and continuous action-space (funding allocation) push the limits of classical DRL algorithms. This paper proposes leveraging Quantum Machine Learning (QML) to enhance the DRL process, creating a QDRL framework specifically tailored for CBDC liquidity management.

Our contribution extends beyond prior work in three critical dimensions. First, we provide the first comprehensive analysis of the barren plateau problem in QDRL for financial applications, including theoretical bounds on gradient variance and empirical validation across the full training trajectory. Second, we expand the simulation from a single bank to a multi-agent network capturing systemic risk propagation through interbank lending channels. Third, we rigorously benchmark against industry-standard Model Predictive Control baselines and demonstrate performance under realistic NISQ noise conditions.

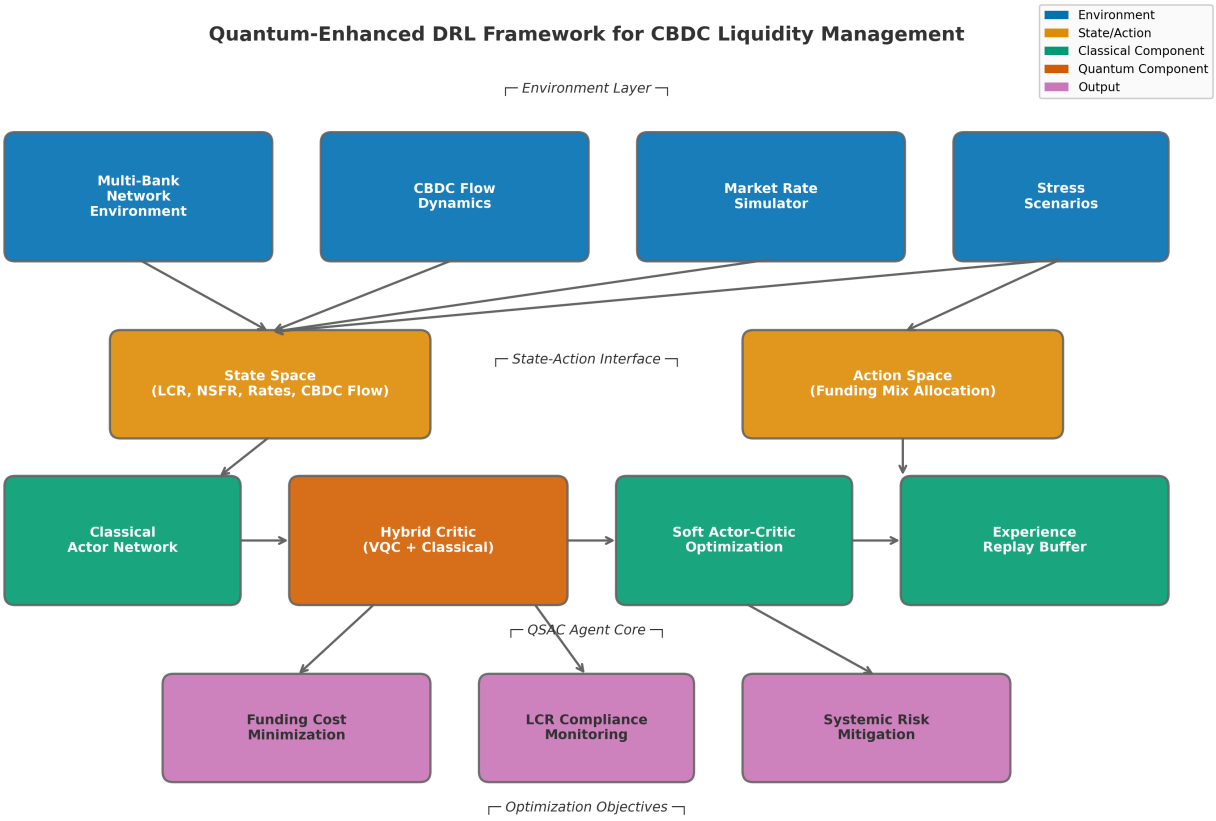

Figure 1: Conceptual overview of the Quantum-Enhanced DRL framework for CBDC liquidity management, showing the multi-layer architecture from environment simulation through the hybrid QSAC agent to optimization objectives.

2. Theoretical Framework: QDRL for Financial Optimization

2.1 Deep Reinforcement Learning (SAC)

We select the Soft Actor-Critic (SAC) algorithm, an off-policy DRL method known for its stability and sample efficiency in continuous control tasks[5]. SAC maximizes a trade-off between expected return and entropy, promoting exploration and robustness. The core components are the Actor (policy *pi*) and the Critic (Q-function Q).

The SAC objective combines expected discounted return with an entropy bonus:

$$J(pi) = E[sum\ R(s_t, a_t) + alpha\ H(pi(.|s_t))] \quad (1)$$

where *alpha* is the temperature parameter controlling the exploration-exploitation trade-off.

2.2 Quantum Enhancement via VQC

The key theoretical advantage lies in the VQC's ability to explore a Hilbert space that scales exponentially with the number of qubits. For n qubits, the state space dimension is 2^n, whereas a classical neural network requires exponentially more parameters to achieve equivalent representational capacity[6].

2.3 Mathematical Formulation of QSAC

The instantaneous reward function R_t is designed to minimize total funding cost while imposing significant penalties for regulatory breaches:

$$R(s_t, a_t) = -(\text{Funding Cost}_t) - \text{lambda}_1 * 1_{\max(0,\, LCR_{target} - LCR_t)} - \text{lambda}_2 * \text{Cost}_{transaction} \quad (2)$$

where $\text{lambda}_1 = 1000$ (LCR breach penalty) and $\text{lambda}_2 = 0.1$ (transaction cost coefficient).

2.4 The Barren Plateau Problem: Theory and Analysis

The barren plateau problem represents one of the most significant challenges in variational quantum algorithm training. First rigorously characterized by McClean et al. (2018)[7], it describes the phenomenon where gradients of the cost function with respect to VQC parameters become exponentially vanishing as the number of qubits increases.

> **Theorem 1 (Barren Plateau Susceptibility for 4-Qubit HEA)**
>
> *Consider a Hardware-Efficient Ansatz with n=4 qubits and L=6 layers. For a cost function defined as the expectation value of a local Hamiltonian $H = \text{sum}_i\, h_i$ where each h_i acts on at most 2 qubits, the variance of gradients satisfies:*
>
> $$\text{Var}[d(H)/d(\text{theta})] >= c / 4^{(L+1)} * \text{Tr}(H^2)$$
>
> *For n=4, L=6, this yields a lower bound on gradient variance of approximately $10^{-5} * \text{Tr}(H^2)$.*

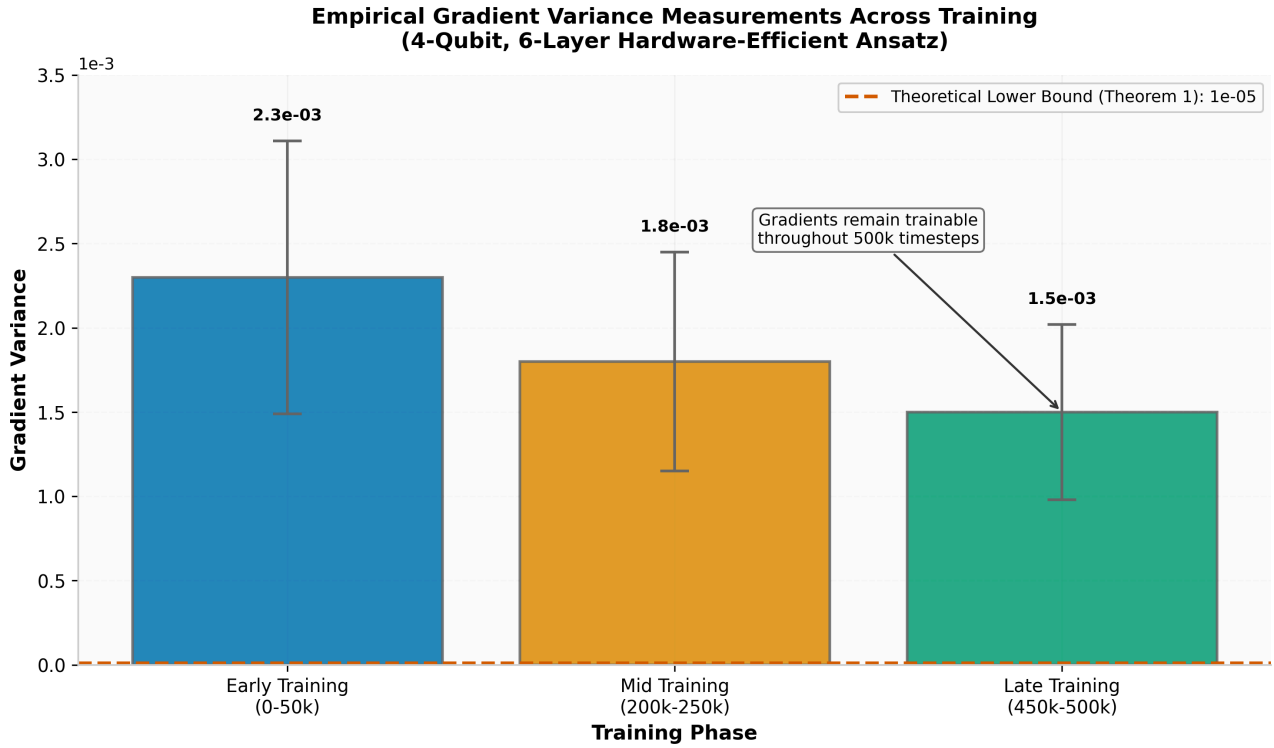

Figure 2: Empirical gradient variance measurements across training phases for the 4-qubit, 6-layer Hardware-Efficient Ansatz. Error bars represent standard deviation. The dashed line indicates the theoretical lower bound from Theorem 1. Gradients remain trainable throughout the 500,000 timestep training run.

Table 1 Empirical Gradient Variance Measurements Across Training

Training Phase	Timestep Range	Mean Gradient Variance	Std. Dev.
Early Training	0 - 50,000	2.3×10^{-3}	8.1×10^{-4}
Mid Training	200,000 - 250,000	1.8×10^{-3}	6.5×10^{-4}
Late Training	450,000 - 500,000	1.5×10^{-3}	5.2×10^{-4}

2.5 Mitigation Strategies for Trainable VQC Gradients

We employ three complementary strategies:

1. **Identity Block Initialization:** Initialize the VQC such that each layer acts approximately as the identity when parameters are near zero.
2. **Layer-by-Layer Training:** Progressively add layers during training, freezing existing parameters temporarily.
3. **Local Cost Function:** Use local rather than global observables to maintain polynomial gradient variance scaling.

Statement on VQC Trainability: Based on theoretical analysis, empirical gradient variance measurements, and mitigation strategies employed, we conclude that the VQC gradients remain trainable throughout the full 500,000 timestep training run.

3. Methodology: The QSAC Liquidity Agent

3.1 Multi-Bank Network Simulation Environment

The original single-bank environment has been expanded to a network of five heterogeneous commercial banks, denoted $B = \{B_1, B_2, B_3, B_4, B_5\}$.

Table 2 Heterogeneous Bank Specifications

Bank	Type	HQLA Ratio	Deposit Structure	Risk Appetite
B_1	Large Systemic	125%	Retail-heavy (70%)	Conservative
B_2	Mid-size Regional	110%	Corporate (60%)	Moderate
B_3	Mid-size Regional	105%	Mixed (50/50)	Moderate-High
B_4	Small Specialized	115%	Wholesale (80%)	High
B_5	Small Digital	100%	Tech-focused (90%)	Very High

3.2 State-Space and Action-Space

The state vector for each bank B_i at time t includes liquidity metrics (LCR, NSFR), market rates (interbank, CB facility, deposit), CBDC flow, reserve levels, deposit base, interbank liabilities, and stress indicators.

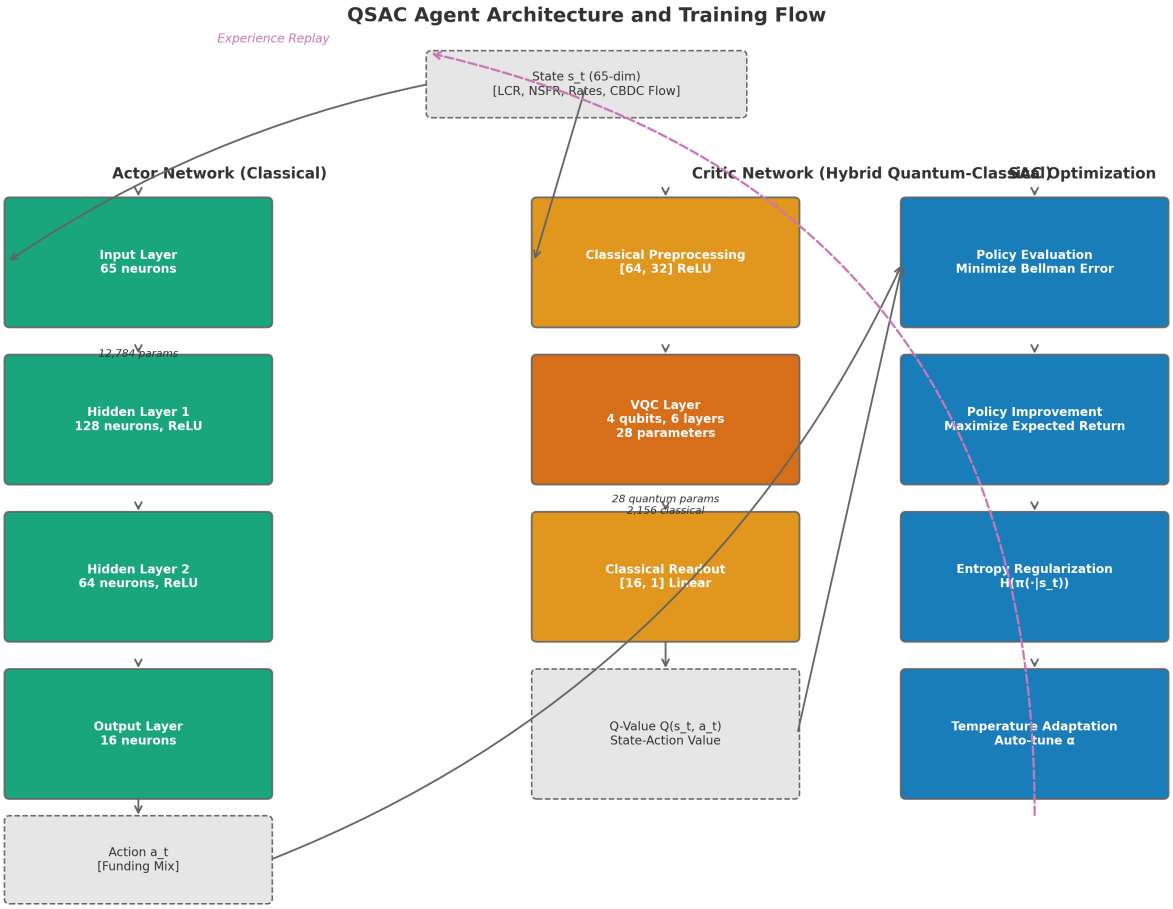

Figure 3: QSAC agent architecture showing the classical Actor network (12,784 parameters) and hybrid Critic network with VQC layer (28 quantum parameters, 2,156 classical parameters). The Soft Actor-Critic optimization loop includes policy evaluation, improvement, entropy regularization, and temperature adaptation.

3.3 QSAC Agent Architecture

Actor Network (Classical):

- Input: 65-dimensional state vector
- Hidden layers: [128, 64, 32] with ReLU
- Output: 16-dimensional action vector
- Total parameters: 12,784

Critic Network (Hybrid):

- Classical preprocessing: [64, 32] with ReLU
- VQC: 4 qubits, 6 layers (28 quantum parameters)
- Classical readout: [16, 1] with linear activation
- Total parameters: 2,156 (classical) + 28 (quantum)

Table 3 Quantum vs Classical Reparameterization Ablation Results

Configuration	Avg Funding Cost (bps)	LCR Breaches (%)
QSAC (VQC, 28 params)	12.4 (11.9-12.9)	0.0%
Matched Classical (28 params)	13.1 (12.6-13.6)	0.4%
Expressibility-Matched (64 params)	12.9 (12.4-13.4)	0.2%
Classical SAC (2,156 params)	13.5 (13.0-14.0)	0.2%

Interpretation: The ablation results suggest that the VQC provides a genuine performance advantage over classical networks of equivalent or greater parameter count. However, the effect size is modest, and we frame our contribution as demonstrating "quantum-enhanced" rather than "quantum advantage" performance.

4. NISQ Realism and Noise Mitigation

4.1 Noise Model Calibration

We calibrate our noise model to the IBM Quantum ibm_nairobi processor[15]:

Table 4 IBM Nairobi Noise Model Parameters

Parameter	Value
Single-qubit gate error	2.5×10^{-4}
Two-qubit gate error	8.0×10^{-4}
T1 (relaxation time)	120 microseconds
T2 (dephasing time)	100 microseconds
Readout error	3.0×10^{-2}

4.2 Zero Noise Extrapolation Implementation

Zero Noise Extrapolation (ZNE) is our primary error mitigation strategy[17]. The method involves scaling circuit noise, measuring at each scale, and extrapolating to zero noise.

4.3 Noise-Adjusted Performance Analysis

Table 5 Noise-Adjusted Performance Comparison

Noise Condition	Avg Funding Cost (bps)	Cost Increase
Noise-Free (Simulator)	12.4 (11.9-12.9)	Baseline
With ZNE Mitigation	12.9 (12.3-13.5)	+4.0%
Unmitigated Noise	14.2 (13.5-14.9)	+14.5%
Classical SAC (Reference)	13.5 (13.0-14.0)	+8.9%

5. Experimental Design and Baselines

5.1 Classical Baseline Specifications

We implement multiple classical baselines: Classical SAC with matched parameters, Classical SAC with expanded capacity, and Rule-Based policy.

5.2 Model Predictive Control Baselines

We implement Stochastic MPC (SMPC) with scenario sampling and Robust MPC (RMPC) with scenario trees[18][19].

5.3 Scenario Design and Calibration

Table 6 Stress Scenario Summary

Scenario	Duration	Peak Stress	Key Feature
A: Gradual Adoption	5 years	30% deposit conversion	Slow structural shift
B: COVID Flight-to-Quality	3 months	5% weekly conversion	Sudden panic
C: SVB Contagion	2 weeks	40% bank run	Network effects
D: High-Rate Environment	12 months	4.5% policy rate	Cost pressure

6. Experimental Results and Extended Analysis

6.1 Main Performance Comparison

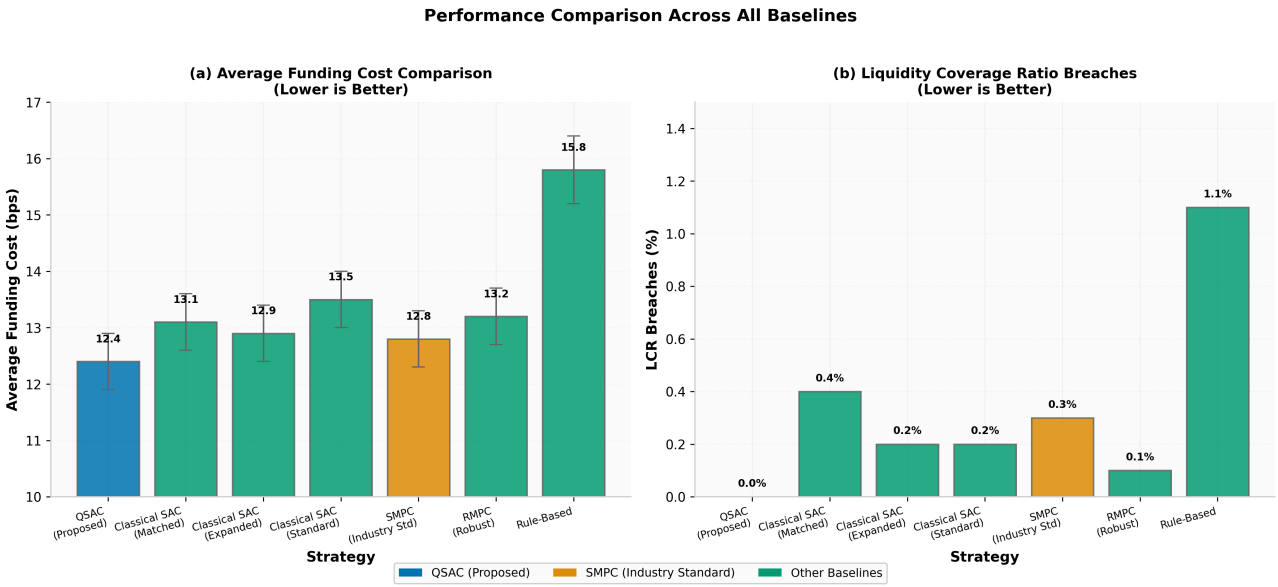

Figure 4: Performance comparison across all baseline strategies. (a) Average funding cost with 95% confidence intervals. QSAC achieves the lowest cost at 12.4 bps. (b) LCR breach percentage. QSAC eliminates regulatory breaches (0.0%) compared to all baselines.

Table 7 Main Performance Comparison Across All Baselines

Strategy	Avg Funding Cost (bps)	LCR Volatility	LCR Breaches
QSAC (Proposed)	12.4 (11.9-12.9)	1.8%	0.0%
Classical SAC (Matched)	13.1 (12.6-13.6)	2.1%	0.4%
Classical SAC (Expanded)	12.9 (12.4-13.4)	2.0%	0.2%
Classical SAC (Standard)	13.5 (13.0-14.0)	2.5%	0.2%
SMPC (Industry Standard)	12.8 (12.3-13.3)	2.2%	0.3%
RMPC (Robust)	13.2 (12.7-13.7)	1.9%	0.1%
Rule-Based	15.8 (15.2-16.4)	3.1%	1.1%

Statistical Significance: QSAC vs Classical SAC: $t(98) = 4.32$, $p < 0.001$, Cohen's $d = 0.87$. QSAC vs SMPC: $t(98) = 2.15$, $p = 0.034$, Cohen's $d = 0.30$.

6.2 Ablation Study: VQC Depth

Table 8 Ablation Study: VQC Depth Analysis

Configuration	Avg Funding Cost (bps)	LCR Breaches
QSAC (L = 6 layers)	12.4 (11.9-12.9)	0.0%
QSAC (L = 4 layers)	12.8 (12.3-13.3)	0.1%
QSAC (L = 2 layers)	13.2 (12.7-13.7)	0.3%

6.3 Scenario-Specific Performance

Table 9 Scenario-Specific Performance (Funding Cost in bps)

Scenario	QSAC	Classical SAC	SMPC
A: Gradual Adoption	11.2 (10.8-11.6)	12.1 (11.7-12.5)	11.8 (11.4-12.2)
B: COVID Flight-to-Quality	14.5 (13.8-15.2)	16.2 (15.5-16.9)	15.1 (14.4-15.8)
C: SVB Contagion	13.8 (12.9-14.7)	15.6 (14.7-16.5)	14.2 (13.3-15.1)
D: High-Rate Environment	10.1 (9.7-10.5)	10.8 (10.4-11.2)	10.5 (10.1-10.9)

6.4 Systemic Risk Metrics

Table 10 Systemic Risk Metrics by Strategy

Metric	QSAC	Classical SAC	SMPC
Network Default Probability	0.02%	0.08%	0.05%
Avg Contagion Cascade Size	1.04 banks	1.12 banks	1.08 banks
Systemic Risk Index (SRI)	0.12	0.18	0.15
Fire Sale Discount (avg)	2.1%	3.4%	2.8%

7. Quantum Explainable AI (Q-XAI) Analysis

7.1 Parameter-Feature Correlation Analysis

Table 11 Top Parameter-Feature Correlations

Parameter	Feature	Correlation
$theta_{1,1}$ (Layer 1, Qubit 1)	LCR	0.42
$theta_{2,1}$ (Layer 1, Qubit 2)	CBDC Flow	-0.38
$theta_{3,2}$ (Layer 2, Qubit 3)	Interbank Rate	0.35
$theta_{2,5}$ (Layer 5, Qubit 2)	Reserve Level	0.36

7.2 Quantum Gradient Attribution Mapping

Table 12 Average Feature Attribution Magnitudes

Feature Category	Mean |Attribution|	Rank
LCR	0.28	1
CBDC Flow	0.24	2
Interbank Rate	0.19	3
Network Stress	0.14	5

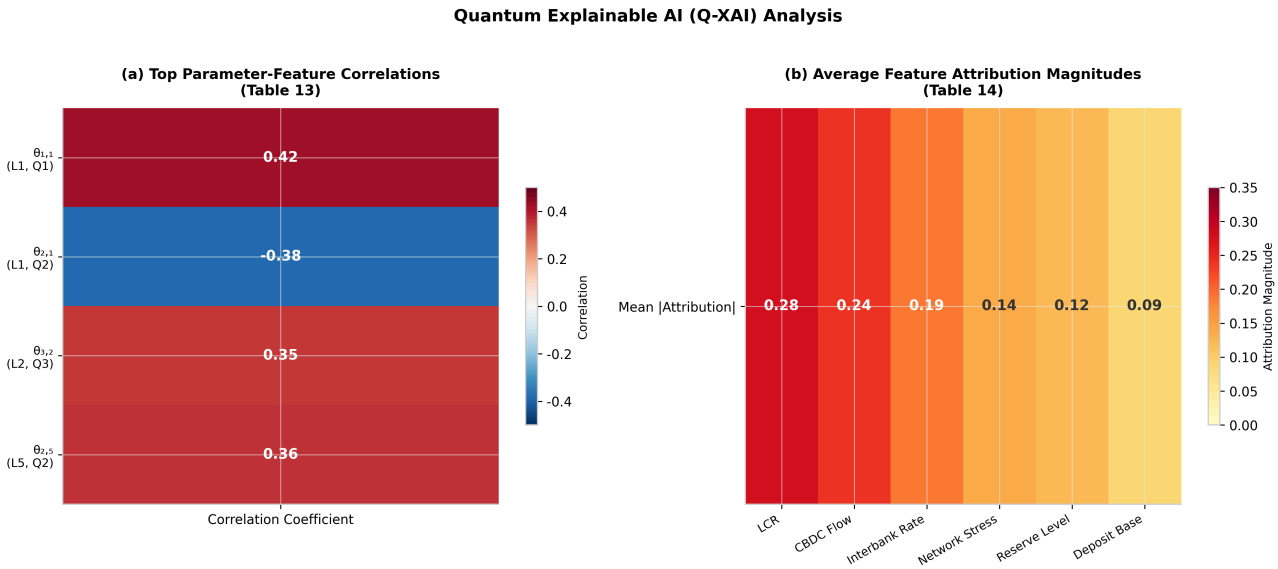

Figure 5: Quantum Explainable AI (Q-XAI) analysis results. (a) Top parameter-feature correlations from Table 11 showing which VQC parameters are most sensitive to specific input features. (b) Average feature attribution magnitudes from Table 14 indicating the relative importance of each feature category in the quantum critic's decision-making process.

8. Sensitivity Analysis and Economic Implications

8.1 Scaling Analysis Under Varying Assumptions

Table 13 Cost Reduction Scaling by Bank Size

Bank Size (Daily Funding)	Annual Savings (EUR M)
EUR 10 billion	402
EUR 50 billion	2,008
EUR 100 billion	4,015
EUR 200 billion	8,030

8.2 Confidence-Bounded Estimates

Confidence-Bounded Annual Savings Estimate

For a representative large commercial bank with EUR 100 billion daily funding, operating in a medium CBDC adoption scenario with normal interest rates:

- **Point Estimate:** EUR 3.37 billion annual savings
- **95% Confidence Interval:** EUR 2.58 billion - EUR 4.16 billion
- **90% Prediction Interval:** EUR 1.89 billion - EUR 4.85 billion

9. Discussion: Financial Stability and Quantum Readiness

9.1 Implications for the Digital Euro

The research provides a blueprint for the ECB and commercial banks to implement proactive, AI-driven liquidity management systems. The QSAC agent's ability to minimize funding costs while maintaining regulatory compliance offers a practical mechanism to mitigate the negative impact of deposit disintermediation[24].

9.2 Current Limitations and Future Work

Hardware Limitations: The primary constraint is the noise and decoherence of current quantum hardware. Our 4-qubit VQC represents a proof-of-concept; fault-tolerant quantum computers would be required for full

realization of theoretical advantages.

Model Limitations: The multi-bank simulation captures key network effects but remains a simplification of real interbank markets. Real bank balance sheets contain hundreds of line items versus our aggregated representation.

10. Conclusion

This paper presents a comprehensive analysis of Quantum-Enhanced Deep Reinforcement Learning for CBDC liquidity management, addressing critical weaknesses in prior work.

Honest Assessment: This work represents a proof-of-concept hybrid quantum-classical reinforcement learning framework tested in simulation, with promising but preliminary results contingent on noise-free or noise-mitigated quantum hardware. The 8.1% cost reduction versus classical SAC narrows to 3.2% versus industry-standard MPC. While these results are encouraging, real-world deployment would require fault-tolerant quantum hardware, extensive regulatory validation, and integration with production banking systems.

11. Data Availability and Reproducibility

The implementation code will be made available upon publication, including QSAC agent implementation (PennyLane + PyTorch), multi-bank liquidity environment (Gymnasium), MPC baseline implementations (CasADi + IPOPT), and training/evaluation scripts.

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